

Security risk assessment report

Part 1: Select up to three hardening tools and methods to implement

Multi-Factor Authentication (MFA)
Strong Password Policy and Access Control
Firewall Configuration and Port Filtering

Part 2: Explain your recommendations

1. Multi-Factor Authentication (MFA)

MFA requires users to provide two or more forms of authentication before accessing systems or data. For example, users may need a password and a one-time code sent to their mobile device.

This technique is effective because even if an attacker obtains a password through password sharing or theft, they still cannot access the account without the second authentication factor. MFA significantly reduces the risk of unauthorized access and data breaches.

Implementation Frequency:

MFA should be enabled permanently for all user and administrator accounts. Authentication systems should be reviewed regularly, at least monthly, to ensure they are functioning correctly.

2. Strong Password Policy and Access Control

The organization should establish a password policy that requires unique, complex passwords, regular password changes, and prohibits password sharing. The default database administrator password should be changed immediately. Additionally, users should only receive the minimum access privileges necessary to perform their job duties (principle of least privilege).

This method is effective because strong passwords are more difficult to crack

through brute-force attacks. Eliminating default passwords removes a common vulnerability that attackers often target. Restricting access privileges limits the damage that can occur if an account becomes compromised.

Implementation Frequency:

Password policies should be enforced continuously. Passwords should be reviewed and updated every 60–90 days, and access privileges should be audited at least quarterly.

3. Firewall Configuration and Port Filtering

The organization's firewalls should be configured with rules that monitor and filter incoming and outgoing network traffic. Only approved ports and services should be allowed, while unnecessary traffic should be blocked.

This technique is effective because firewalls act as a barrier between trusted internal networks and external threats. Port filtering reduces the attack surface by preventing unauthorized connections and blocking malicious traffic before it reaches critical systems.

Implementation Frequency:

Firewall rules should be monitored continuously and reviewed monthly. Rules should also be updated immediately after any security incident or when network changes occur.